

Functions of many variables – 1

Partial derivatives: a survey of the subject

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Theoretical minimum.

Definition. f is *differentiable* at x if $f(x+h) = f(x) + \langle a, h \rangle + o(\|h\|)$ for some vector a , and then $a = \nabla f(x)$ is the gradient. Differentiability implies the existence of all directional derivatives, and $\partial_v f(x) = \langle \nabla f(x), v \rangle$; the converse fails, and the existence of all partial derivatives does not even imply continuity. What is true is one implication only: if all partial derivatives exist near x and are continuous at x , then f is differentiable at x .

Theorem 1 (chain rule and the mean value theorem). For differentiable f and a differentiable path γ one has $(f \circ \gamma)'(t) = \langle \nabla f(\gamma(t)), \gamma'(t) \rangle$. Applying this to a segment gives, for a real-valued f ,

$$f(b) - f(a) = \langle \nabla f(a + \theta(b-a)), b-a \rangle \quad \text{for some } \theta \in (0,1).$$

For a map into $\mathbb{R}^m$ with $m \geq 2$ no such equality holds; only the inequality $\|f(b) - f(a)\| \leq \sup \|Df\| \cdot \|b-a\|$ survives (Functions of many variables – 2).

Theorem 2 (Schwarz). If $\partial_x \partial_y f$ and $\partial_y \partial_x f$ exist near a point and are continuous at it, they are equal there. Continuity is essential: for $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ (and $f(0,0) = 0$) the two mixed derivatives at the origin are -1 and $+1$.

Theorem 3 (Euler's identity). A C^1 function on $\mathbb{R}^n \setminus \{0\}$ satisfies $f(tx) = t^k f(x)$ for all $t > 0$ if and only if $\langle x, \nabla f(x) \rangle = k f(x)$. The operator $E = x_1 \partial_1 + \dots + x_n \partial_n$ measures the degree, and comparing powers of t in $f(tx)$ separates a polynomial into homogeneous parts. This is the engine behind problems 1 and 8 of the homework; the topic is continued in Functions of many variables – 4.

Theorem 4 (closed versus exact). If $f_1, \dots, f_n$ are C^1 on a convex (or merely simply connected) domain and $\partial_j f_i = \partial_i f_j$ for all i, j , then $f_i = \partial_i g$ for a single function g . On a domain with a hole this is false: the field $(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2})$ is closed and has no potential. The whole circle of questions about potentials, circulation and flux — and with it the two Putnam problems that used to stand here — is developed in Multiple integrals – 4.

Fact 1. The Laplacian $\Delta f = \partial_1^2 f + \dots + \partial_n^2 f$; a polynomial with $\Delta P = 0$ is called harmonic, and the homogeneous parts of a harmonic polynomial are harmonic again. Hadamard's lemma: if f is C^1 and $f(0) = 0$, then $f(x) = \sum x_i g_i(x)$ with continuous g_i ; this is the statement of the last homework problem and the standard way to divide by a coordinate.

Fact 2 (where each line continues). The gradient as an instrument — Functions of many variables – 2; maps of $\mathbb{R}^n$ into itself, the Jacobian and invertibility — Functions of many variables – 3; homogeneity and functional equations in several variables — Functions of many variables – 4. Convex functions on $\mathbb{R}^n$ live in the series Convex – 1...10, constrained extrema in Differentiable functions – 7 and Optimization – 1...4, and harmonic functions of two variables in Complex analysis – 8.

1. (Putnam 1961) For a real-valued function $f(x, y)$ of the two positive real variables x and y , define $f(x, y)$ to be linearly bounded if and only if there exists a positive number K such that $|f(x, y)| < (x + y)K$ for all positive x and y . Find necessary and sufficient conditions on the real numbers α and β such that $x^\alpha y^\beta$ is linearly bounded.

2. (Putnam 1967) Let f be a real-valued function having partial derivatives and which is defined for $x^2 + y^2 \leq 1$ and is such that $|f(x, y)| \leq 1$. Show that there exists a point (x_0, y_0) in the interior of the unit circle such that

$$\left(\frac{\partial f}{\partial x}(x_0, y_0)\right)^2 + \left(\frac{\partial f}{\partial y}(x_0, y_0)\right)^2 \leq 16.$$

3. (Капачёв 1.56) Does there exist a function of two variables, continuous

on the whole plane, which has exactly two local extrema and for which the value at the point of local minimum is greater than the value at the point of local maximum?

4. (Mexmat, 2009) Let $f: \mathbb{R}^2 \rightarrow [0, \infty)$ be continuous and suppose that for some $C > 0$, every unit square K and every $x \in K$ one has $f(x) \leq C + C \iint_K f$. Prove that $f(x) \leq Me^{C\|x\|}$ for some $M \geq 0$.
5. (Putnam 1999) Let $A = \{(x, y) : 0 \leq x, y < 1\}$. For $(x, y) \in A$, let

$$S(x, y) = \sum_{\frac{1}{2} \leq \frac{m}{n} \leq 2} x^m y^n,$$

where the sum ranges over all pairs (m, n) of positive integers satisfying the indicated inequalities. Evaluate $\lim_{(x,y) \rightarrow (1,1), (x,y) \in A} (1-xy^2)(1-x^2y)S(x, y)$.